

Two-Stage Miller CMOS Op-Amp — Design & Simulation Guide

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Cadence Virtuoso IC23.1 · SkyWater SKY130 PDK · 1.8 V

Target Specifications

Spec	Target	Spec	Target
DC open-loop gain	≥ 65 dB	Load capacitance C_L	5 pF
Phase margin	$\geq 60^\circ$	Supply VDD	1.8 V
Gain-bandwidth (GBW)	≈ 8 MHz	Bias current (tail)	20 μ A
Slew rate	≥ 10 V/ μ s	Comp. cap C_c	1.5 pF

Topology

A classic two-stage Miller-compensated op-amp: an NMOS differential input pair (M1, M2) with a PMOS current-mirror active load (M3, M4) forms the high-gain first stage; a PMOS common-source transistor (M6) with an NMOS current-source load (M7) forms the second gain stage. A Miller capacitor C_c with a nulling resistor R_z ($\approx 1/g_{m6}$) between the output and the first-stage output node (A) sets the dominant pole and cancels the right-half-plane zero for stability. M5 is the tail current source; M8 and the reference current mirror the bias to M5 and M7.

Two-Stage Miller CMOS Op-Amp — SKY130 (1.8 V) | NMOS input pair · PMOS mirror load · PMOS 2nd stage

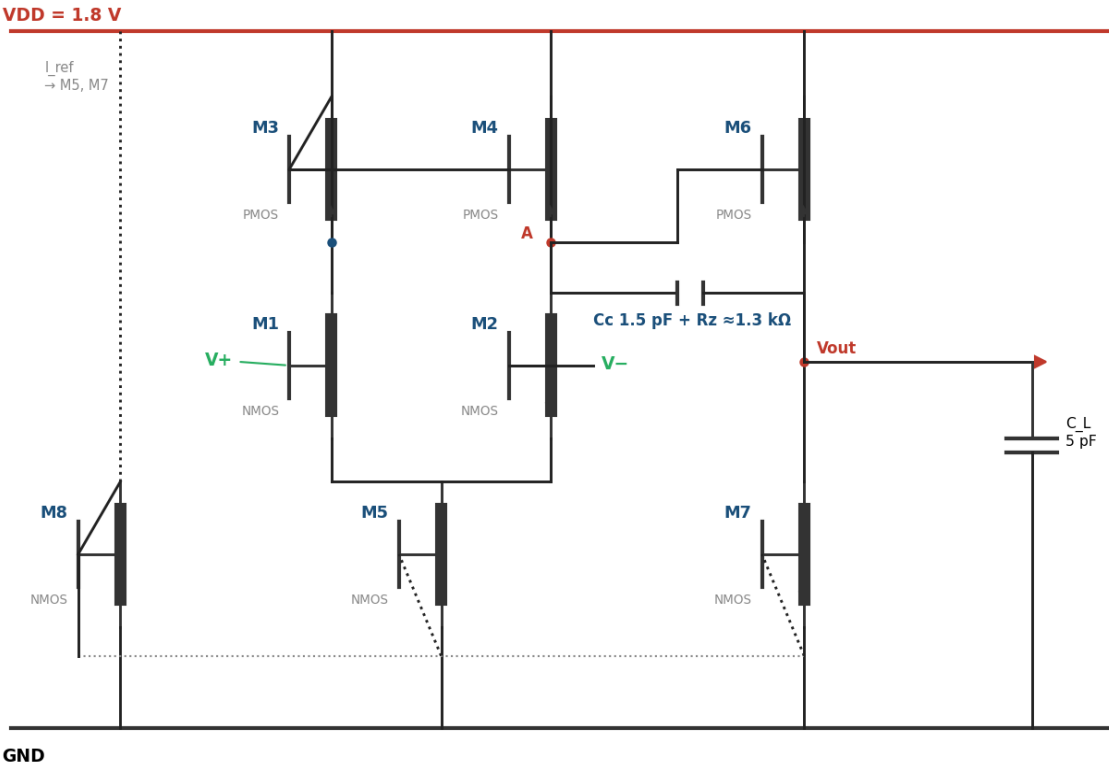


Figure 1. Two-stage op-amp topology (arrows on sources denote device type).

Device Sizing (starting point — tune in simulation)

Device	Type	Role	Drain current	Start W / L
M1, M2	NMOS	Input differential pair	10 μ A each	2 μ m / 0.5 μ m
M3, M4	PMOS	Current-mirror load	10 μ A each	6 μ m / 0.5 μ m
M5	NMOS	Tail current sink	20 μ A	4 μ m / 0.5 μ m
M6	PMOS	2nd-stage common source	75 μ A	45 μ m / 0.5 μ m
M7	NMOS	2nd-stage load	75 μ A	15 μ m / 0.5 μ m
M8	NMOS	Bias mirror (diode)	20 μ A (I_{ref})	4 μ m / 0.5 μ m

Devices: sky130_fd_pr__nfet_01v8 (NMOS) and sky130_fd_pr__pfet_01v8 (PMOS). $L = 0.5 \mu\text{m}$ (well above minimum) improves gain and matching. $C_c = 1.5 \text{ pF}$, $R_z \approx 1.3 \text{ k}\Omega$. Connect all NMOS bodies to GND and PMOS bodies to VDD. Tune each W in simulation to hit the target g_m while keeping every device saturated ($V_{DS} > V_{DSAT}$).

Build Order in Virtuoso

- In your library, **File** → **New** → **Cellview** → opamp_2stage, view *schematic*.
- Instantiate the six device types (bindkey **i**), set W/L per the table, wire per Figure 1. Add C_c and R_z from analogLib (cap, res).
- Add pins (bindkey **p**): v_{inp} , v_{inn} , v_{out} , v_{dd} , v_{ss} , i_{bias} . Check & Save (bindkey **X**) — fix any errors.
- Create a **symbol** (Create → Cellview → From Cellview) for use in the test bench.
- Make a second cell opamp_tb (schematic): drop the symbol, add $v_{dd}=1.8 \text{ V}$, an $i_{dc}=20 \mu\text{A}$ into i_{bias} , and the $C_L=5 \text{ pF}$ load on v_{out} .

ADE Simulation Recipe (Spectre)

- DC operating point** — set input common-mode $\approx 0.9 \text{ V}$. Run, then *Results* → *Print* → *DC Operating Points*: confirm every transistor has $V_{DS} > V_{DSAT}$ (in saturation) and read g_m/I_D . Adjust W to hit $g_{m1} \approx 75 \mu\text{S}$ and $g_{m6} \approx 750 \mu\text{S}$.
- AC — gain, GBW, phase margin**. Easiest is Spectre **stb** (stability) analysis: insert an *iprobe* (analogLib) in the feedback loop of a unity-gain-connected test bench, run stb 1 Hz–10 GHz. Read **DC gain** (low-freq magnitude), **GBW** (0 dB crossing), and **phase margin** directly. (Alternative: open-loop AC using a 1 GHz feedback inductor + 1 GF cap to set DC bias while breaking the loop for AC.)
- Transient — slew rate**. Wire the op-amp as a unity-gain buffer ($v_{out} \rightarrow v_{inn}$), drive v_{inp} with a 0.9 V→1.4 V step (rise $\approx 1 \text{ ns}$), and measure dV_{out}/dt on the rising and falling edges.
- Extra credit**: input common-mode range & output swing (DC sweep), CMRR and PSRR (AC), and total quiescent power ($\approx V_{DD} \times \text{total current}$).

Then: Layout & Verification

Once the schematic meets spec, build the layout (Layout XL), run **DRC** and **LVS** (Calibre or PVS) until clean, extract parasitics with **Quantus QRC**, and re-run the AC/transient tests **post-layout** to confirm the specs hold with real parasitics. Capture the gain/phase plot, the slew waveform, and the DRC/LVS-clean summary for the portfolio.